Gabriele La Malfa

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While studying for my master's degree in accounting, I got hugely passionate about game theory. As I continued my business studies, I focused on moving toward science. I learned programming, mathematics, and statistics on my own, which helped me gain admission to the PhD program in computer science at King's College London. There, I was able to fully pursue my passion for science. Now, in my final year of doctoral studies, I am researching multi-agent reinforcement learning, as well as the safety and fairness of AI. Lately, my work has focused on self-play, large language models, the Theory of Space, and vision-language models.
I am a human-oriented person. Projects are exciting, but people around me and collaborators always come first.

ACADEMIC POSITIONS

PhD candidate in Safe & Trusted AI at King's College London Oct. 2022 – Current
Safety and Fairness in Multi-Agent Systems.
– Supervisors: Prof. Elizabeth Black, Dr Jie M. Zhang and Prof. Michael Luck.

Visiting Researcher at Nokia Bell Labs Jul. 2025 – Dec 2025
Classification of AI agent risks in the workplace.

Research Assistant at the University of Cambridge, Centre for Risk Studies Oct. 2020 – Aug. 2022
Research and modelling business, financial and cyber risks.

WORK EXPERIENCE

Quantitative Risk Officer Jun. 2020 – Aug. 2020
Deutsche Börse AG

EDUCATION

EMLyon Business School 2020
MSc Quantitative Finance

University of Eastern Piedmont 2017
Master's Degree in Accounting and Business Administration

PUBLICATIONS

The Attacker in the Mirror: Breaking Self-Consistency in Safety via Anchored Bipolicy Self-Play.
The Fortieth Annual Conference on Neural Information Processing Systems. Main Track. **[Oral]**
Gabriele La Malfa, Emanuele La Malfa, Saar Cohen, Jie M. Zhang, Michael Luck, Michael Wooldridge, Elizabeth Black.

Reflections and Fragments: Securing LLMs Against Sequential Mosaic Attacks.
Under submission.
Gabriele La Malfa, Emanuele La Malfa, Saar Cohen, Mickel Liu, Christian Schroeder de Witt, Natasha Jaques, Michael J. Wooldridge.

Explore, Map, Remember, Decide: Are Embodied VLMs Ready for Safety-Critical Scenarios?
Under submission.
Gabriele La Malfa, Nitay Alon, Emanuele La Malfa, Reuth Mirsky, Stefan Sarkadi.

Unaccountable Delegation, Fading Skills: Mapping the Risks of Workplace AI Agents.
Proceedings of the Ninth AAAI/ACM Conference on AI, Ethics, and Society (AIES-26) - Main Track. **[Oral]**
Gabriele La Malfa, Lakmal Meegahapola, Edyta Bogucka, Jie M. Zhang, Michael Luck, Elizabeth Black, Daniele Quercia.

Fairness Aware Reinforcement Learning via Proximal Policy Optimization.
Vol. 8 No. 1 (2025): Proceedings of the AAAI Conference on Artificial Intelligence, 40(27), 22725–22733. **[Oral]**
Gabriele La Malfa, Jie M. Zhang, Michael Luck, Elizabeth Black.

Large Language Models Miss the Multi-Agent Mark.
The Thirty-Ninth Annual Conference on Neural Information Processing Systems. Position Paper Track.
Gabriele La Malfa, Emanuele La Malfa, Samuele Marro, Jie M. Zhang, Elizabeth Black, Michael Luck, Philip Torr, Michael Wooldridge.

Using Protected Attributes to Consider Fairness in Multi-Agent Systems.
Proceedings of the AEQUITAS 2024 ECAI Workshop on Fairness and Bias in AI (2024).
Gabriele La Malfa, Jie M. Zhang, Michael Luck, Elizabeth Black.

Deep Neural Networks via Complex Network Theory: a Perspective.
Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence Main Track. Pages 4361-4369.
Gabriele La Malfa, Emanuele La Malfa, Giuseppe Nicosia, Vito Latora.

CURRENT/FUTURE PROJECTS

Language-driven geometric cognition in VLMs. I investigate whether VLMs exhibit human-like geometric inductive biases. Through 2D lexical framing tasks and 3D spatial scaling tests in realistic scenarios, this project evaluates how specific vocabulary and spatial prepositions dynamically alter a model's spatial reasoning and environmental navigation.

Multi-agent Theory of Space in safety-critical scenarios. Building on my paper "Explore, Map, Remember, Decide: Are Embodied VLMs Ready for Safety-Critical Scenarios?", I am exploring extending the work to further safety-critical scenarios in the presence of deception of focal points and multi-agent coordination.

Self-play red teaming for autonomous agents Operative Systems (OS) security. I extend the concept of single-agent oversight to an AI agent "anti-malware" system, in which a population of AI agents (blue team) monitors the health of an OS across both user and agent spaces to detect, intercept, and shield against faulty or malicious behaviours of other AI agents (red team). Through lightweight, distributed Parameter-Efficient Fine-Tuning (PEFT), this framework trains attackers and defenders in a self-reinforcing loop, aiming at secure and safe deployment.

INVITED TALKS & PANELS

Panelist: AI at Work: Power and Social Justice 2026
AAAI/ACM Conference on AI, Ethics, and Society (AIES-26)

Fairness in Multi-Agent Systems Sept. 2023
Machine Learning and Prediction Modelling Seminars

TEACHING

Artificial Intelligence 2023–2025
Practical

Mathematics and Statistics for AI 2024–2025
Group Tutorial Lead

Machine Learning 2023–2024
Group Tutorial Lead

Logic and Knowledge Representation 2023–2024
Group Tutorial Lead

Elementary Logic and Applications 2023–2024
Group Tutorial Lead

Big Data Technology 2023–2024
Group Tutorial Lead

SERVICE & ORGANISATIONS

LOD Conference Series (2020–2025)
Organizing Committee

AAAI, NeurIPS
Reviewer (Top Reviewer for NeurIPS 2026)

VOLUNTEERING

I am committed to social justice and humanitarian support for students and people who are not as lucky as I am.

Lunchtime Dialogues: Identity & Social Justice 2023 – 2025
Lead Facilitator: Organised a dialogue series across all KCL faculties to raise awareness on social justice and human rights.

Ukrainian Global University 2022 – 2023
Interviewer: Supported the international placement program by conducting admissions interviews for Ukrainian students seeking to study at partner universities abroad.

The Kharkiv and Przemyśl Project August 2022
Volunteer: Provided on-the-ground support in Przemyśl, Poland, assisting refugees arriving from and returning to Ukraine.

Centro Gazza Ladra 2018
Project Organiser: Managed crowdfunding and event logistics for an association dedicated to supporting children with special needs.